Ethanol Blending in Gasoline – Czechia

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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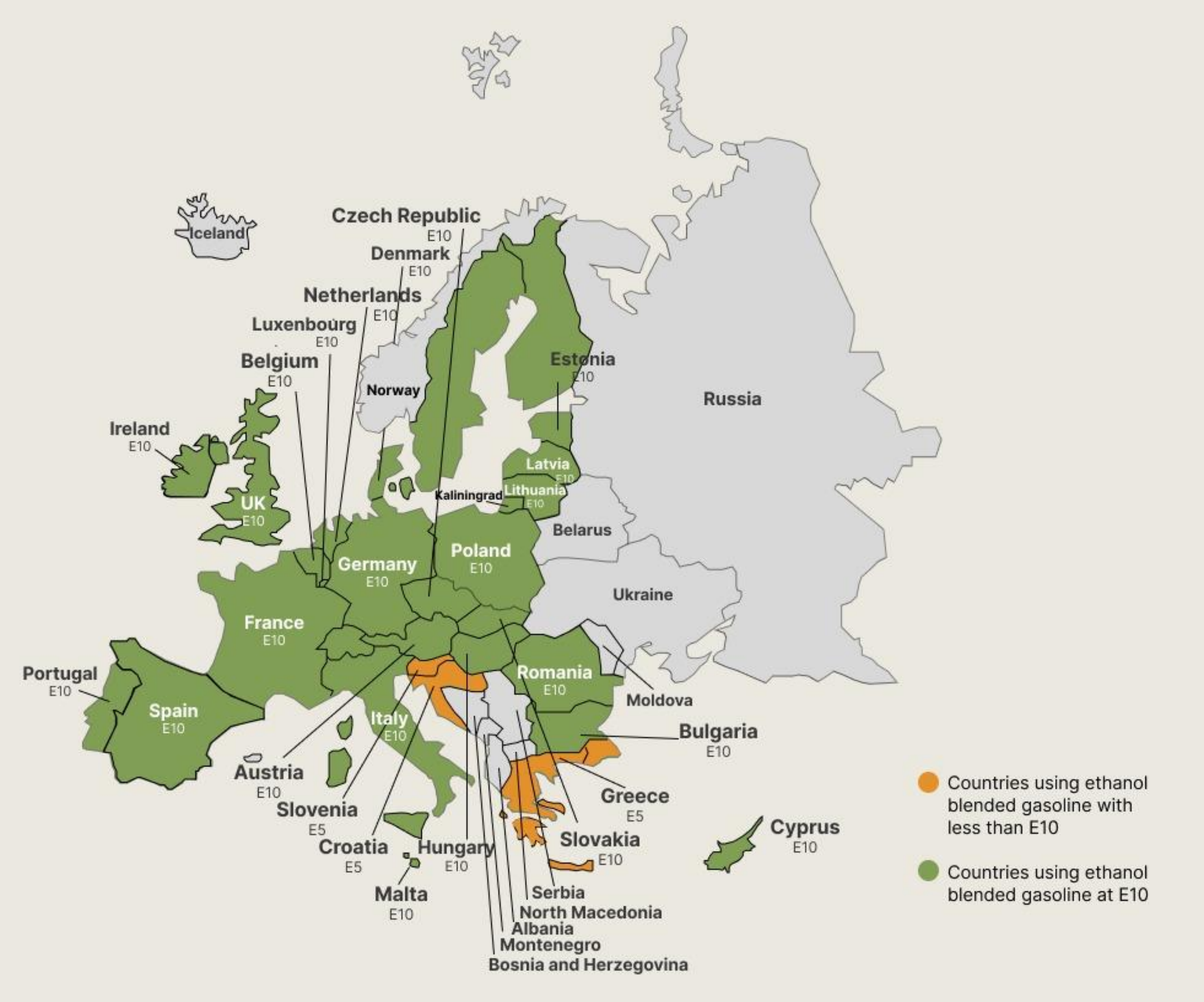
Ethanol Blending in Europe

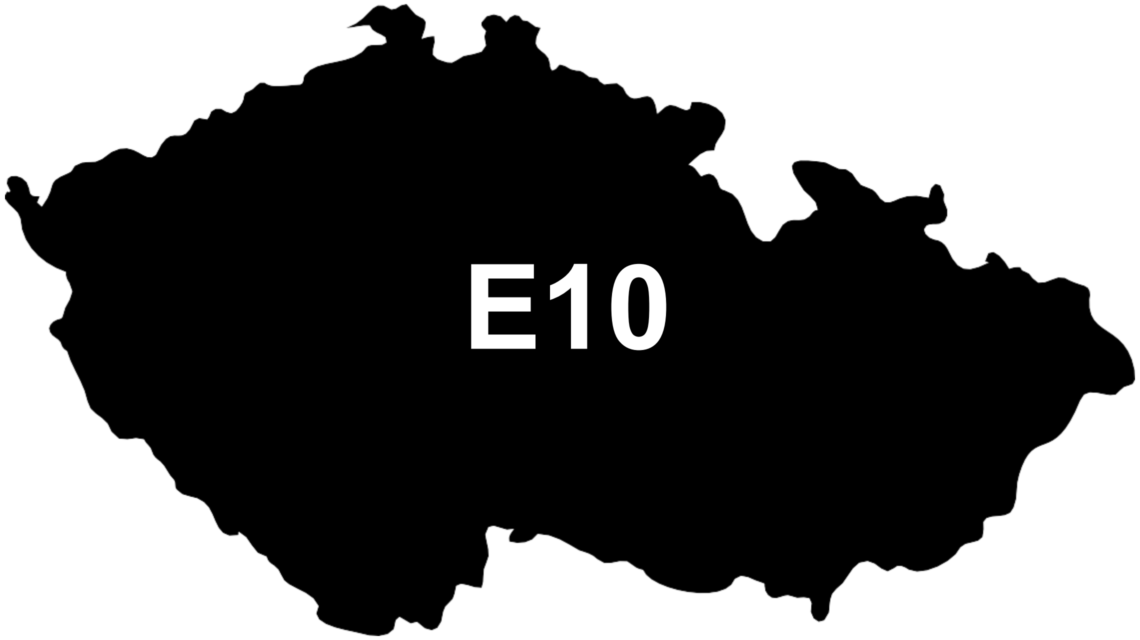
There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Czechia

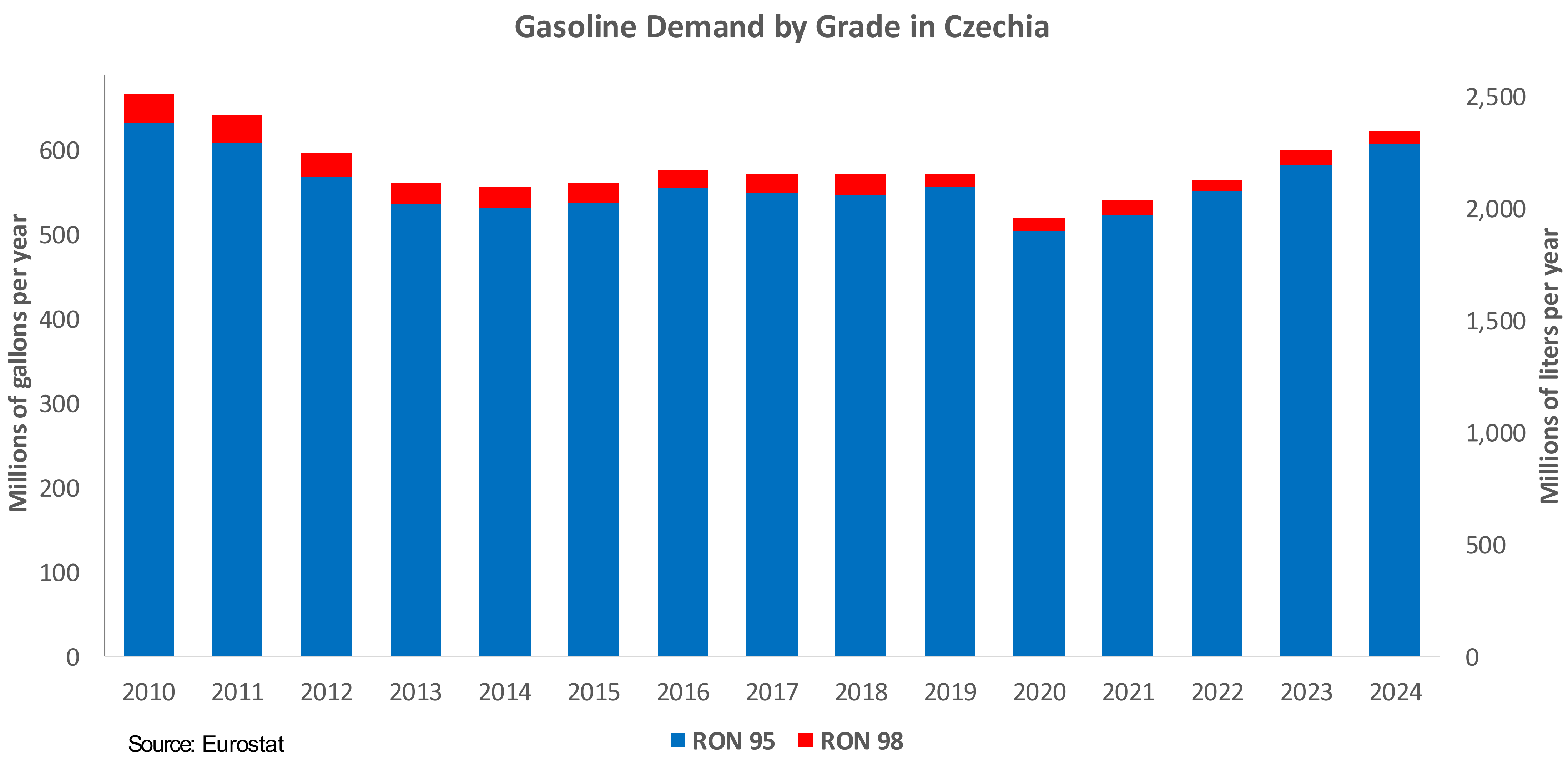


In 2024, gasoline consumption in Czechia was 2,347 million liters (620 million gallons) and growth rate was 1.7%. Gasoline production and net imports were 2,328 and -936 million liters (615 and -247 million gallons) correspondingly. Czechia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.1 RON.

In 2024, ethanol consumption in Czechia was 96 million liters (25 million gallons) representing 4.1% of gasoline demand. Ethanol production and Net Imports were 76 and 21 million liters (25 and 5 million gallons) correspondingly. Ethanol sources are Corn 50% and Sugar Beet 50%.

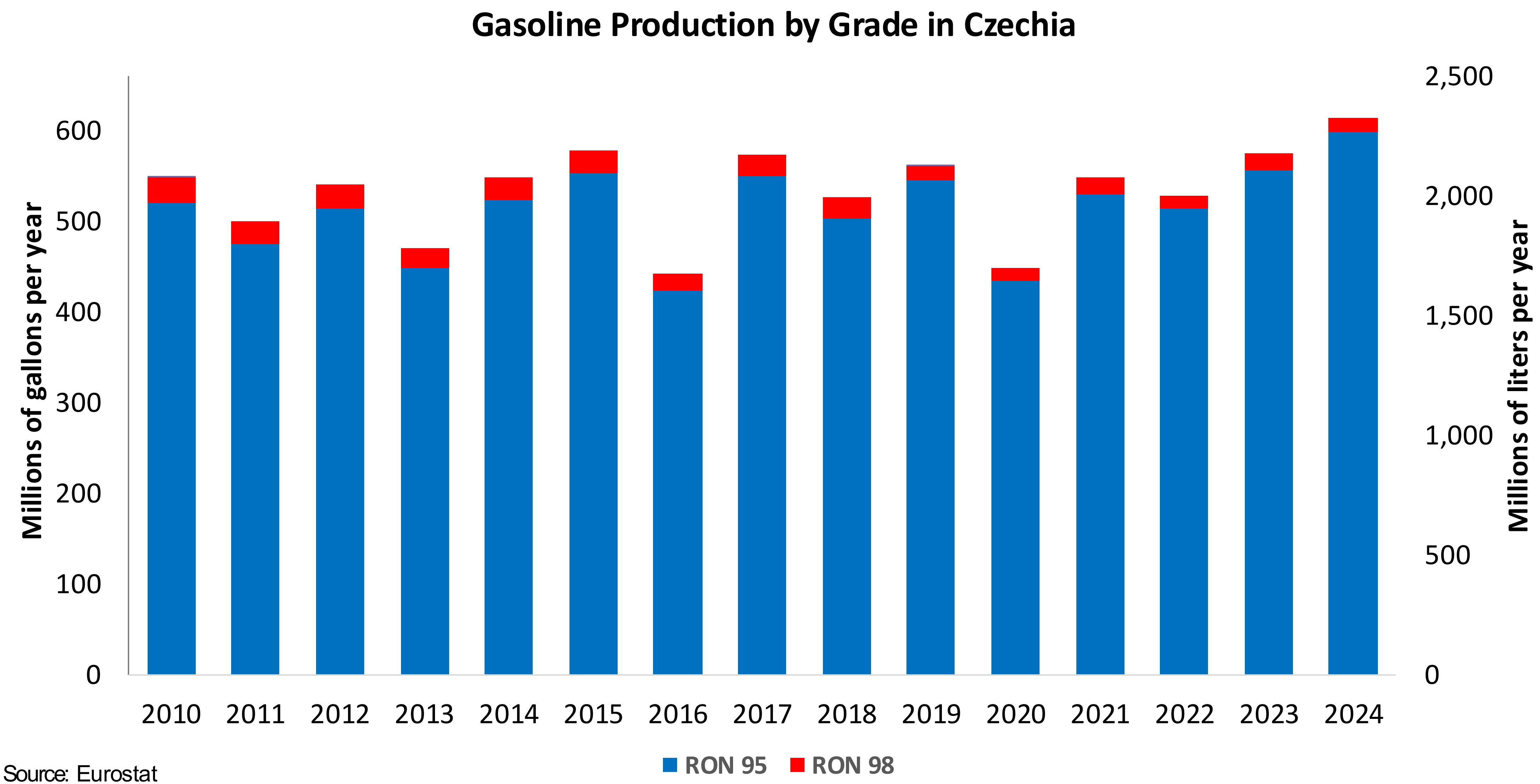
Source: Czechia Statistics, Eurostat

Gasoline Demand in Czechia



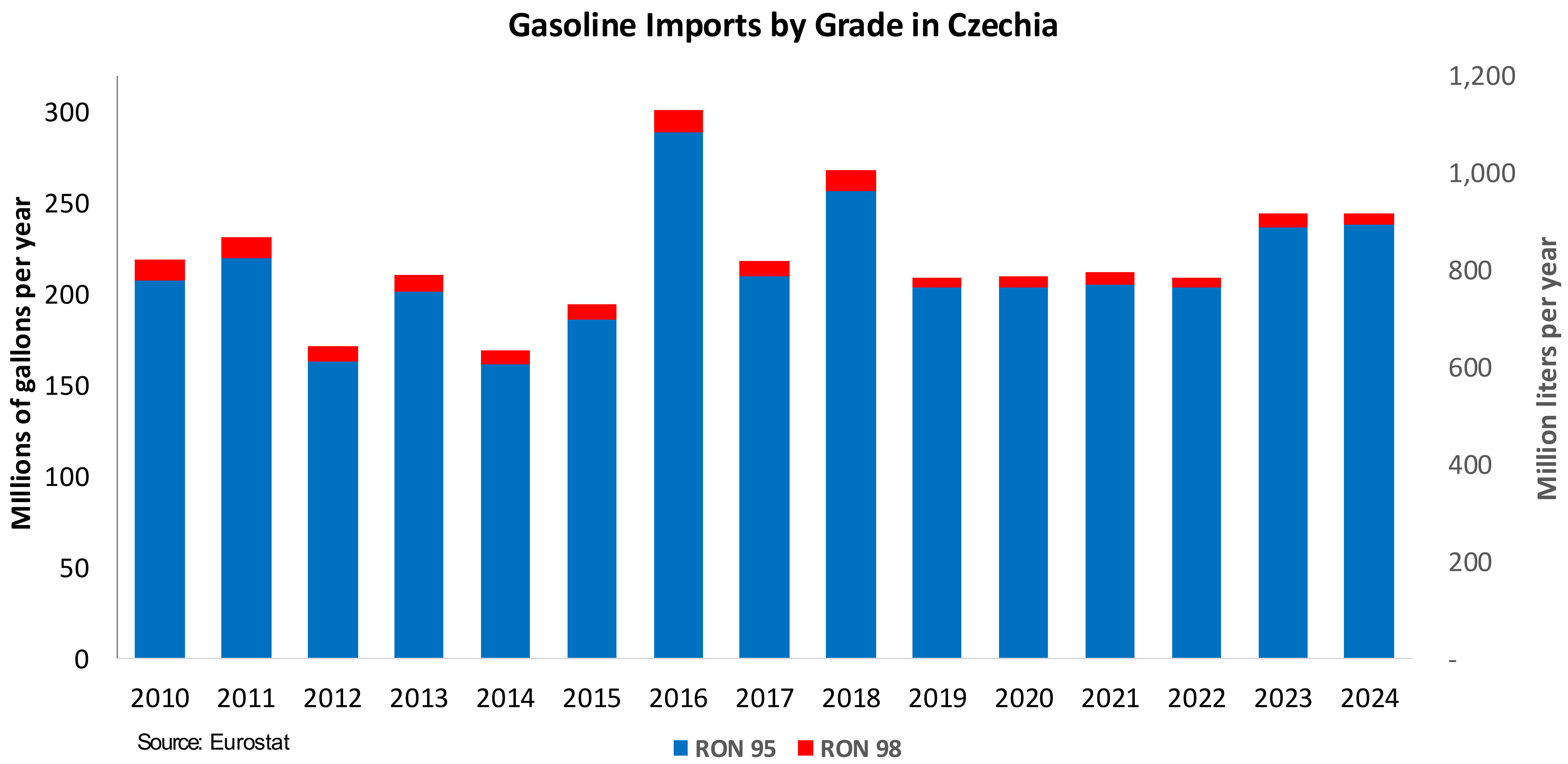
Source: Eurostat

Gasoline Production in Czechia



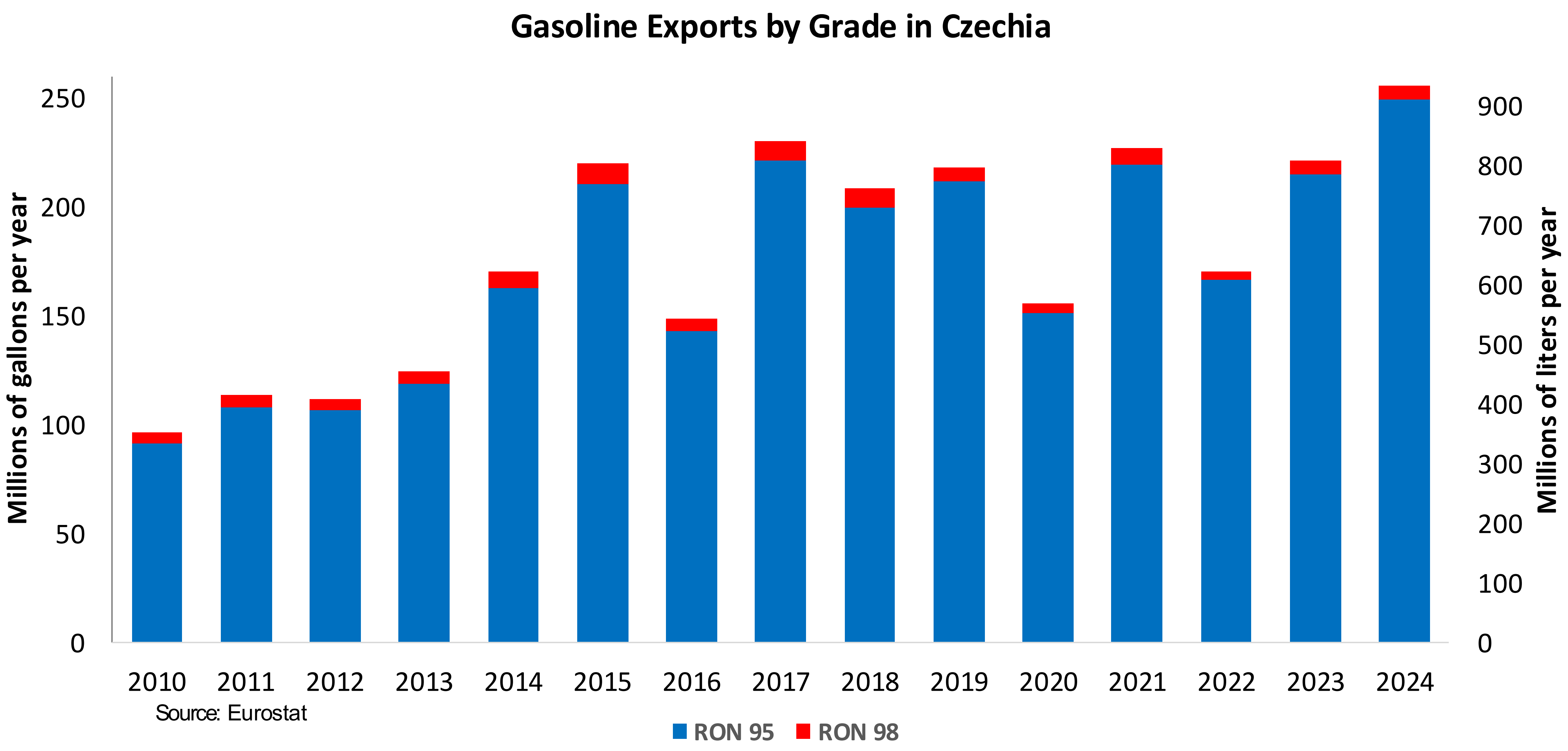
Source: Eurostat

Gasoline Imports in Czechia

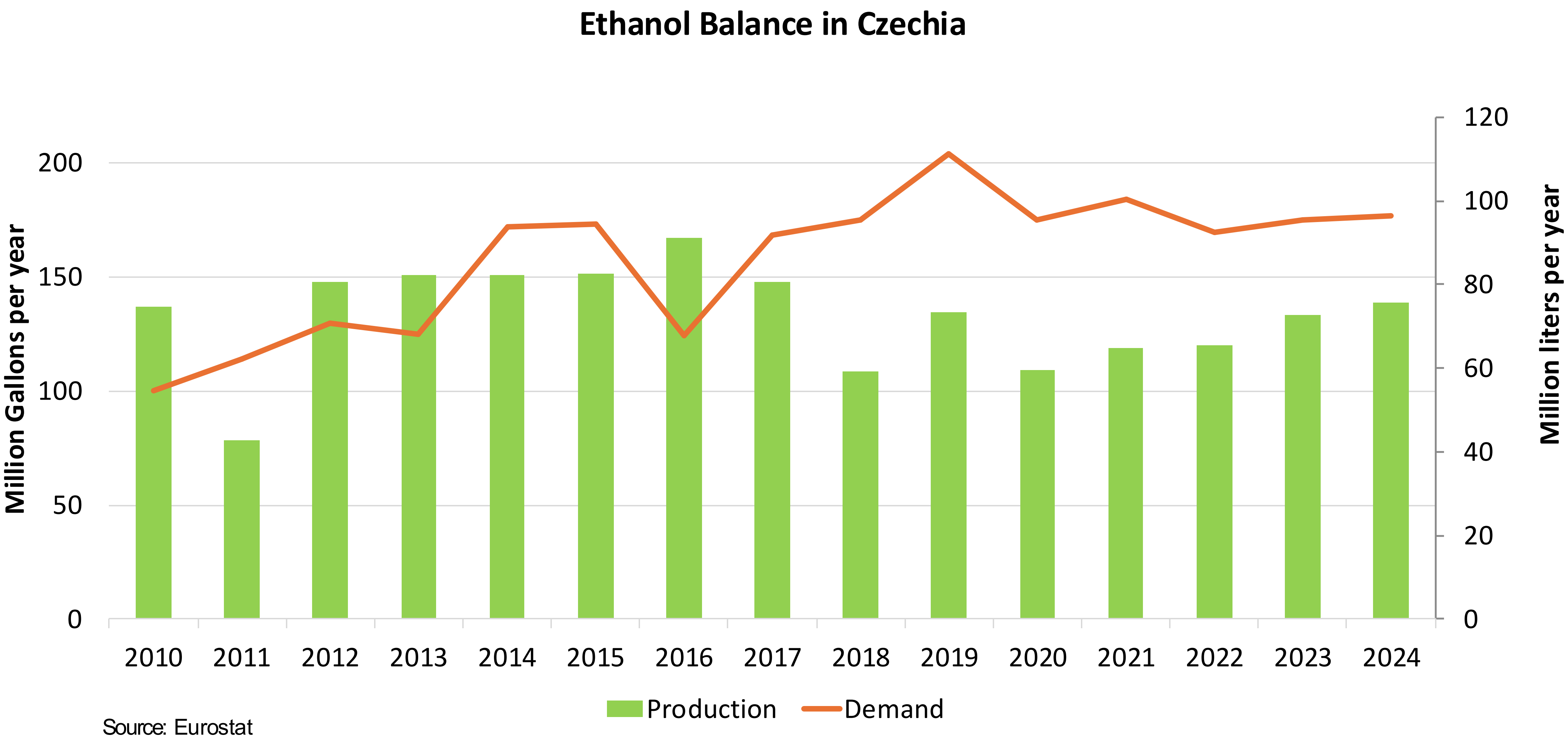


Source: Eurostat

Gasoline Exports in Czechia



Ethanol Balance in Czechia



Source: Eurostat

Gasoline Quality in Czechia

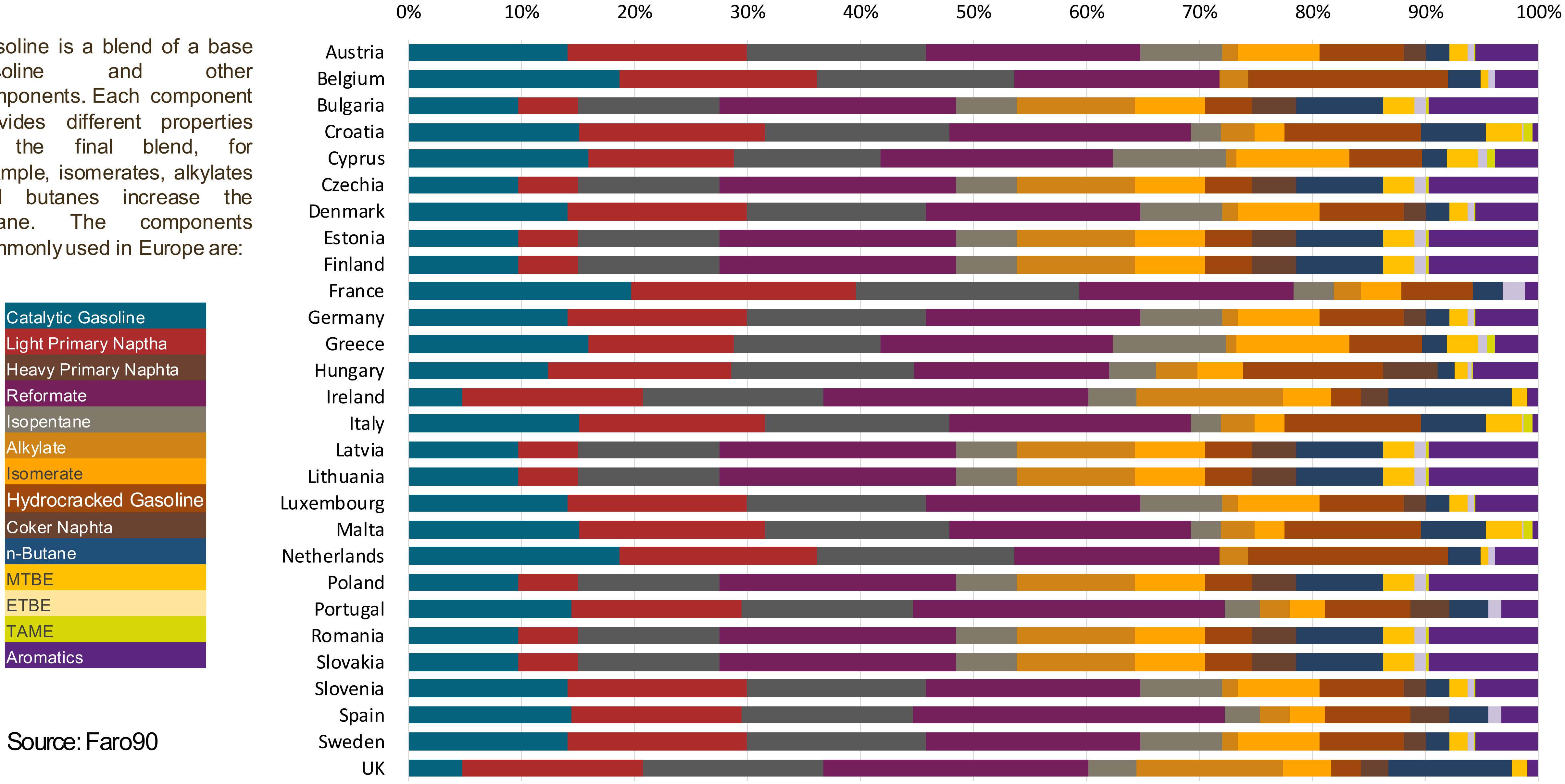
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

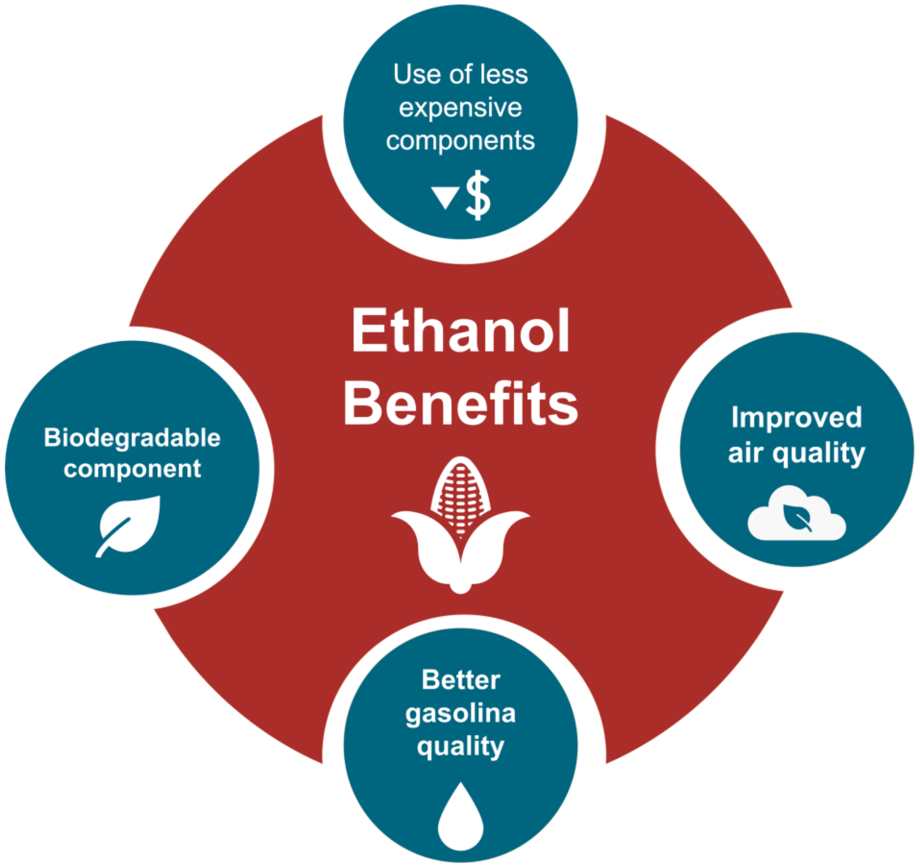
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

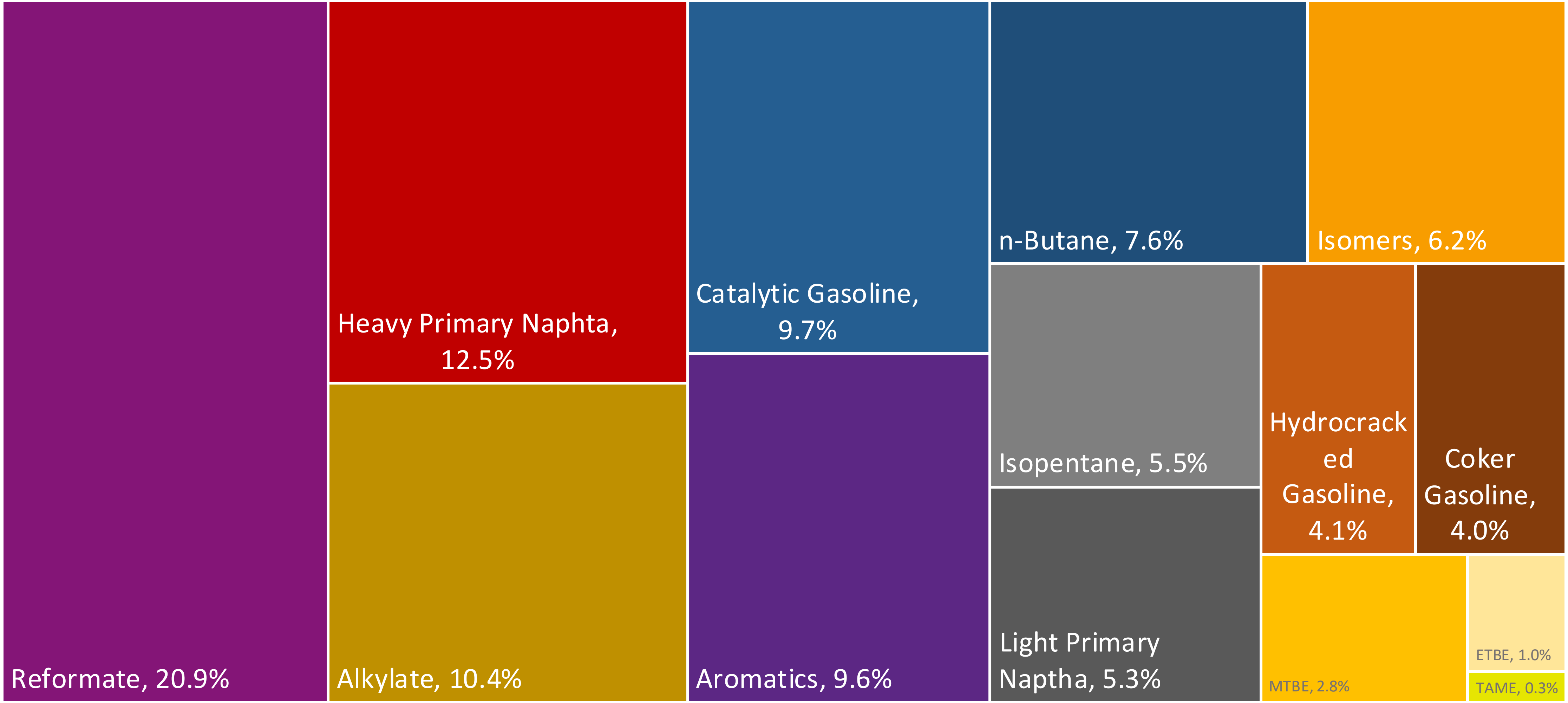
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

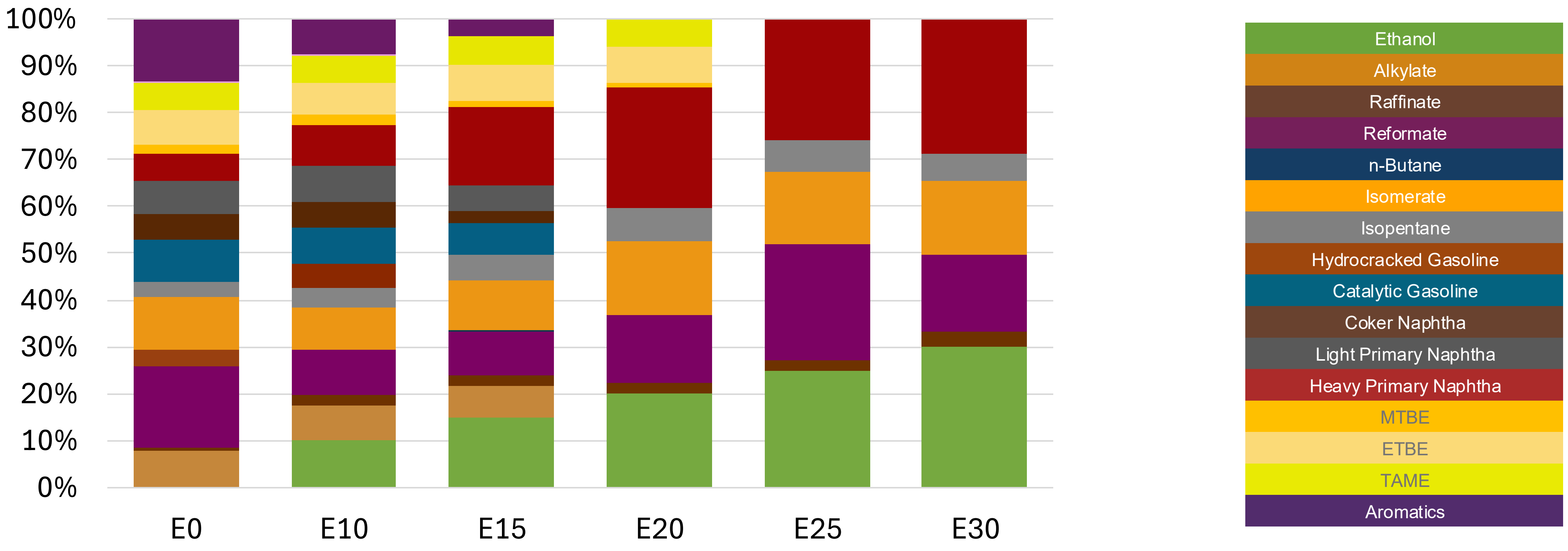


Czechia Available Blending Components



Source: Faro90

Czechia – Gasoline 95 – Constant Octane

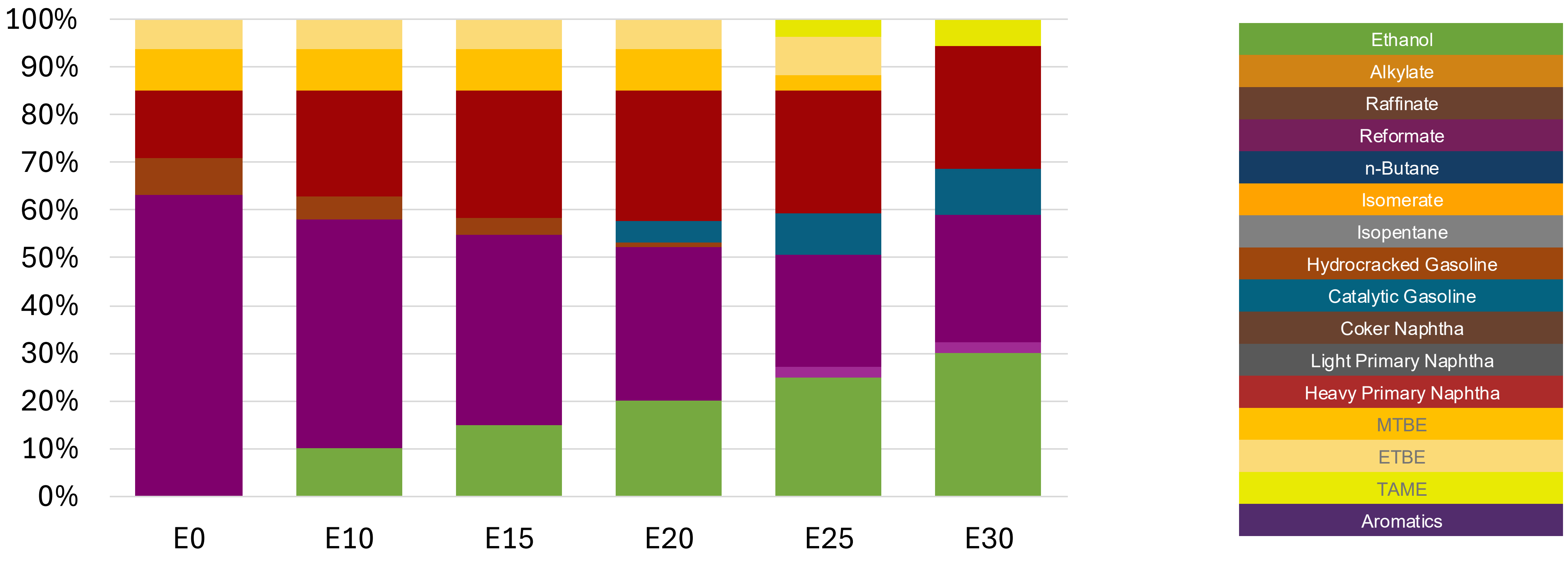


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.314	$ 2.176	$ 2.076	$ 1.961	$ 1.915	$ 1.862

Average CIF 2024 Prices

Source: Faro90

Czechia - Gasoline 98 - Constant Octane

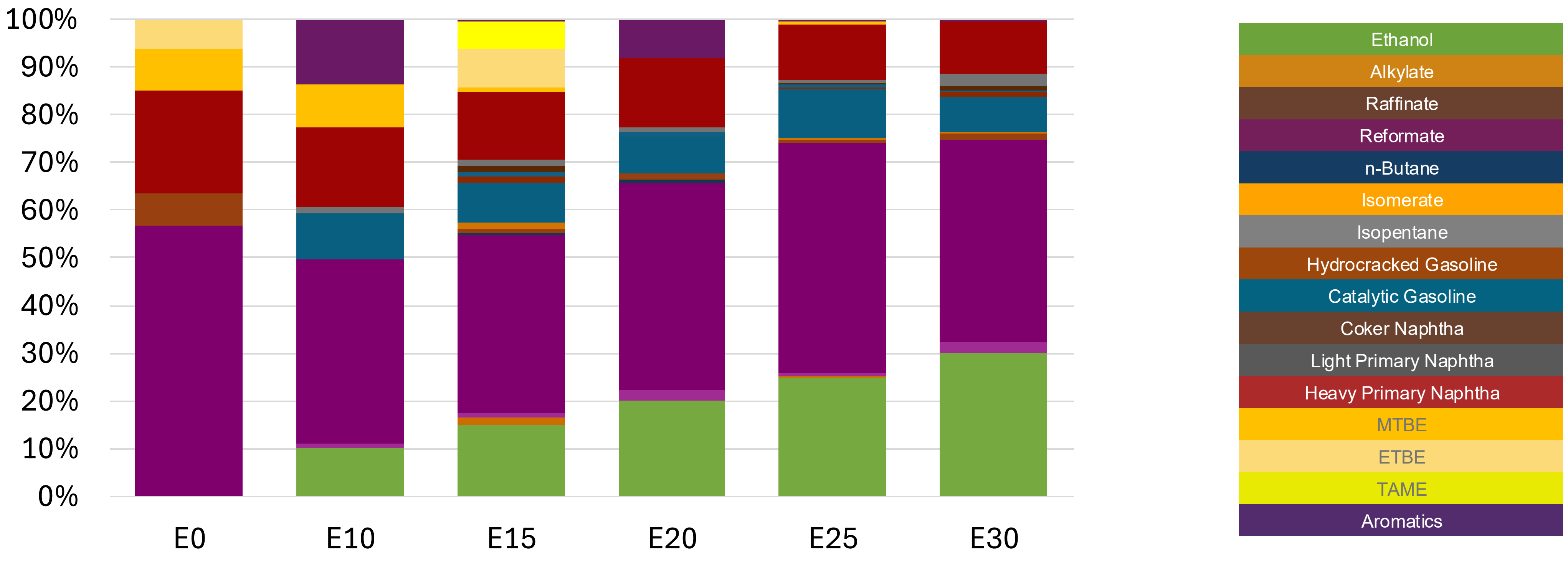


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.334	$ 2.217	$ 2.148	$ 2.076	$ 2.011	$ 1.961

Average CIF 2024 Prices

Source: Faro90

Czechia – Gasoline 95 – Increased Octane

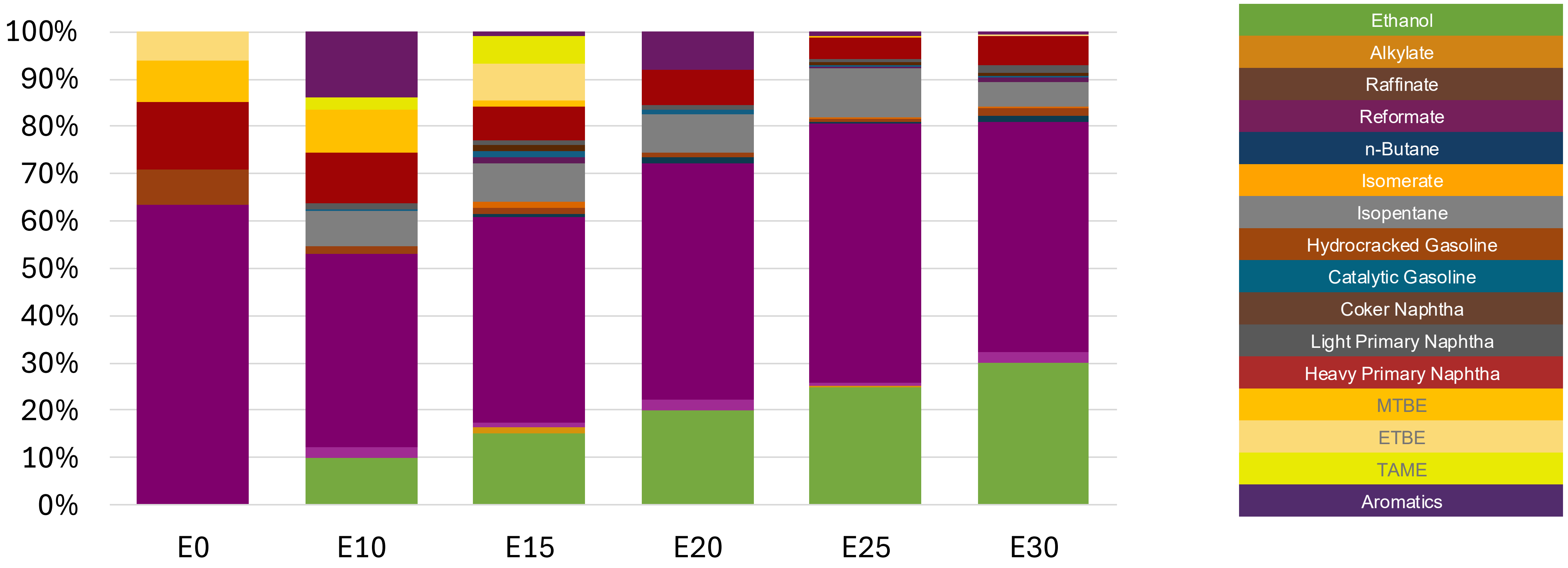


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.202	$ 2.202	$ 2.202	$ 2.202	$ 2.202	$ 2.202

Average CIF 2024 Prices

Source: Faro90

Czechia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.334	$ 2.334	$ 2.334	$ 2.334	$ 2.334	$ 2.334

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

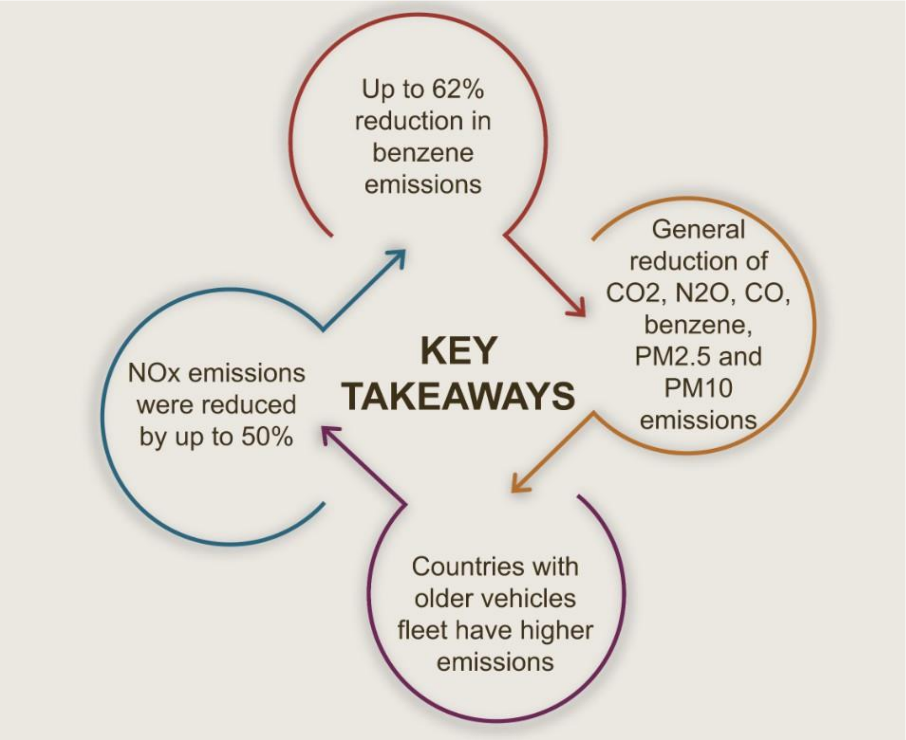
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Czechia

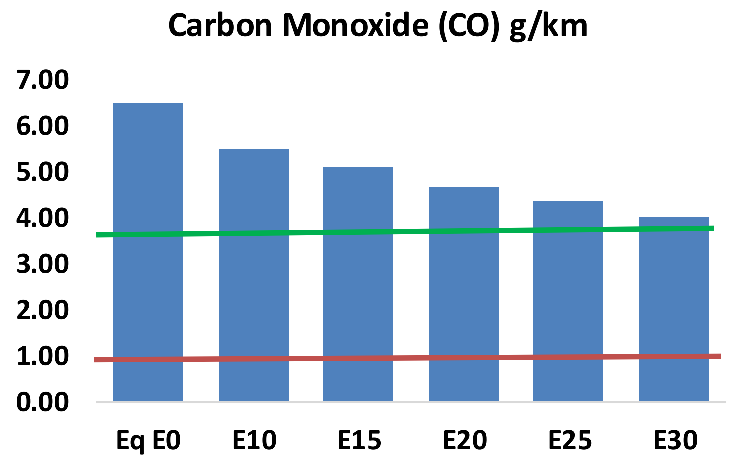
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	79,074
		Single FI	3-Ways		>8 years	496,911
		Multi FI	3-Ways		>8 years	577,810
			Euro III		>8 years	387,206
			Euro IV		>8 years	539,299
			Euro V		>8 years	200,096
			Euro 6		>8 years	444,886
					4-8 years	598,016
					<4 years	712,246
			Hybrid		All	27,035
		Electric			All	73,149
	Medium	Carburator	3-Ways	PCV Tank	>8 years	48,378
		Single FI	3-Ways		>8 years	304,010
		Multi FI	3-Ways		>8 years	353,503
			Euro III		>8 years	236,892
			Euro IV		>8 years	329,942
			Euro V		>8 years	122,419
			Euro 6		>8 years	272,181
					4-8 years	365,866
					<4 years	435,751
			Hybrid		All	16,540
		Electric			All	73,149
Sml Engines	Light	4-Cycle	None	PCV	>8 years	380,304
			Euro III		>8 years	127,130
			Euro IV		>8 years	177,066
			Euro V		>8 years	65,697
			Euro 6		>8 years	146,068
					4-8 years	196,345
					<4 years	233,849
			Hybrid		All	8,662
		Electric			All	23,436
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	207,008
		Fuel-Injection	3-Ways		>8 years	196,071
			Euro III		>8 years	127,335
			Euro IV		>8 years	172,946
			Euro V		>8 years	63,028
			Euro 6		>8 years	138,369
					4-8 years	182,578
					<4 years	225,872
			Hybrid		All	-
		Electric			All	742

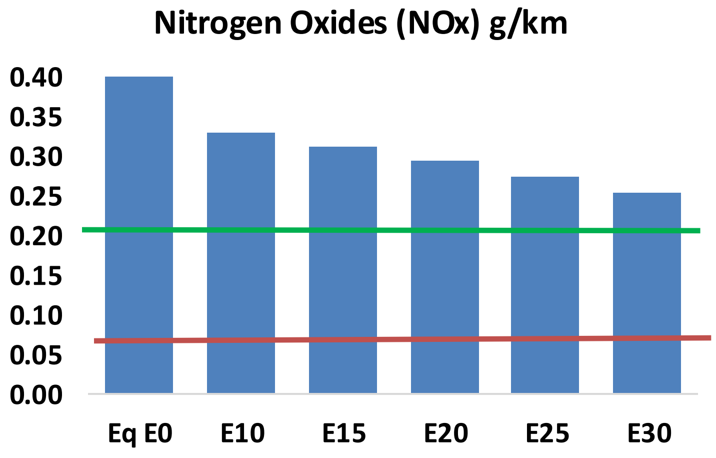
Vehicle Fleet: 9,366,865
Gasoline Vehicle Fleet: 3,468,160

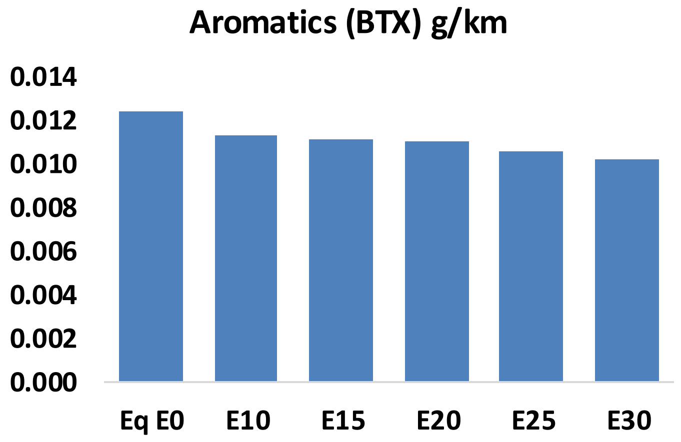
Source: Eurostat, ACEA

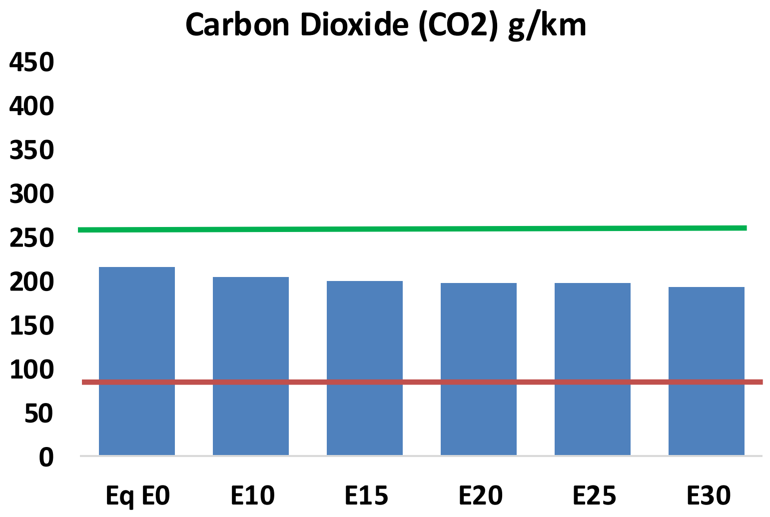
Czechia – Vehicular Emissions

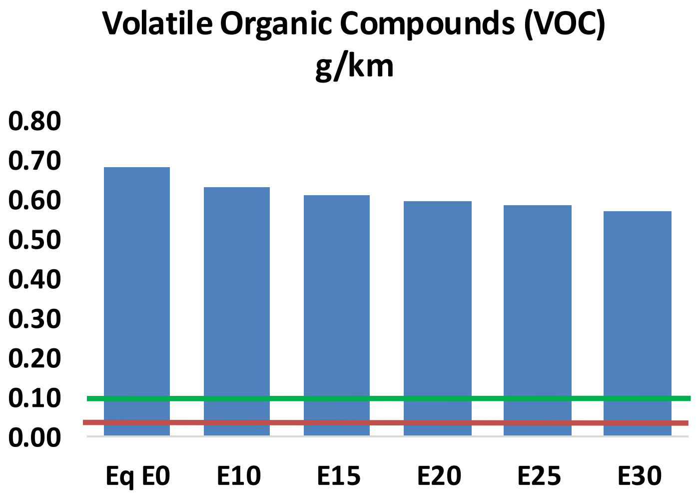
TIER III BIN 125 United States
Euro 6 Standard

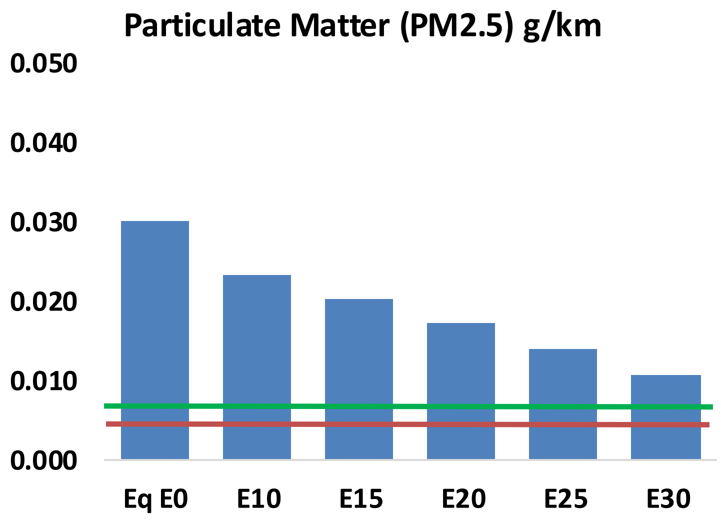












Source: Faro90

Czechia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	514,816	475,239	435,661	403,229	370,591	345,495	316,625	-15%	-28%
CO2	17,030,957	16,593,018	16,179,409	15,854,118	15,693,151	15,618,037	15,330,151	-5%	-8%
VOC	54,073	51,964	49,856	48,528	47,375	46,544	45,240	-8%	-12%
NOx	37,565	28,173	26,295	24,783	23,373	21,817	20,172	-30%	-38%
SOx	196	181	166	154	142	129	115	-15%	-28%
NH3	5,868	5,810	5,752	5,779	5,845	5,890	5,928	-2%	0%
N2O	308	306	305	306	313	316	320	-1%	2%
CH4	11,875	11,875	11,875	12,053	12,314	12,540	12,754	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	361	343	324	310	297	288	274	-10%	-18%
Acetaldehyde	771	885	1,299	1,977	2,694	3,079	3,638	68%	249%
Formaldehyde	2,546	2,475	2,885	3,388	3,549	3,926	4,274	13%	39%
Benzene	987	948	898	884	877	839	812	-9%	-11%
PM2.5	1,023	909	794	689	585	475	360	-22%	-43%
PM10	1,353	1,202	1,050	912	774	628	476	-22%	-43%

Source: Faro90

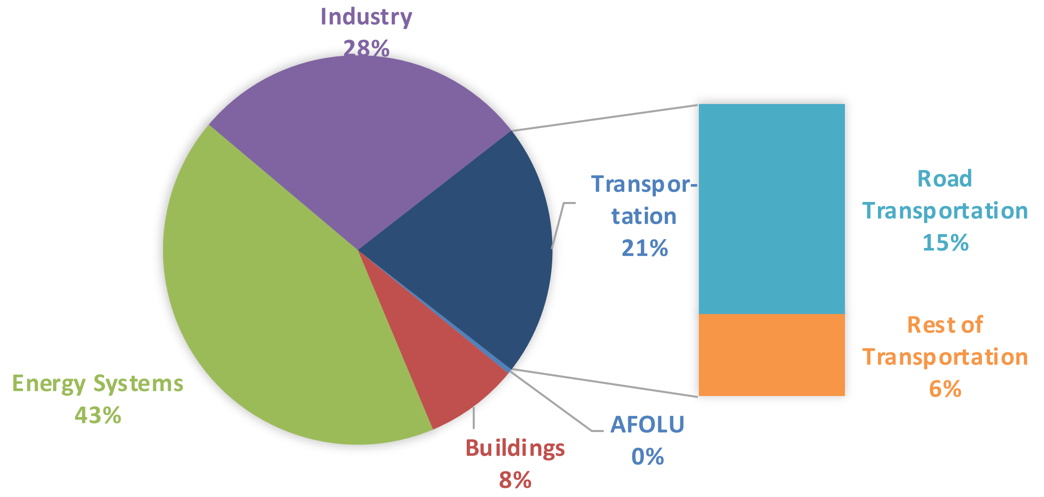
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

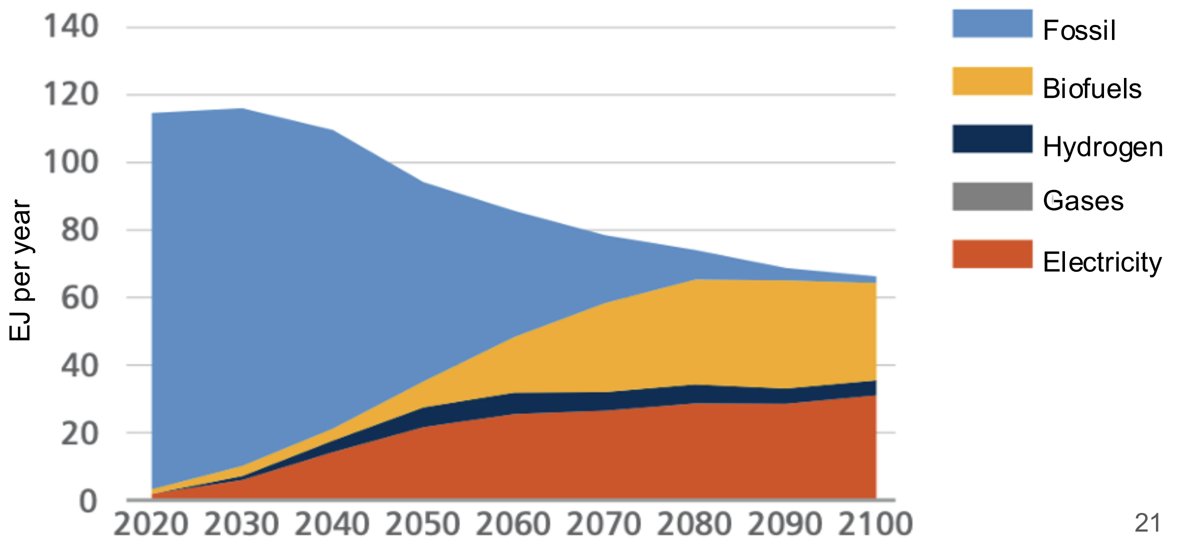
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

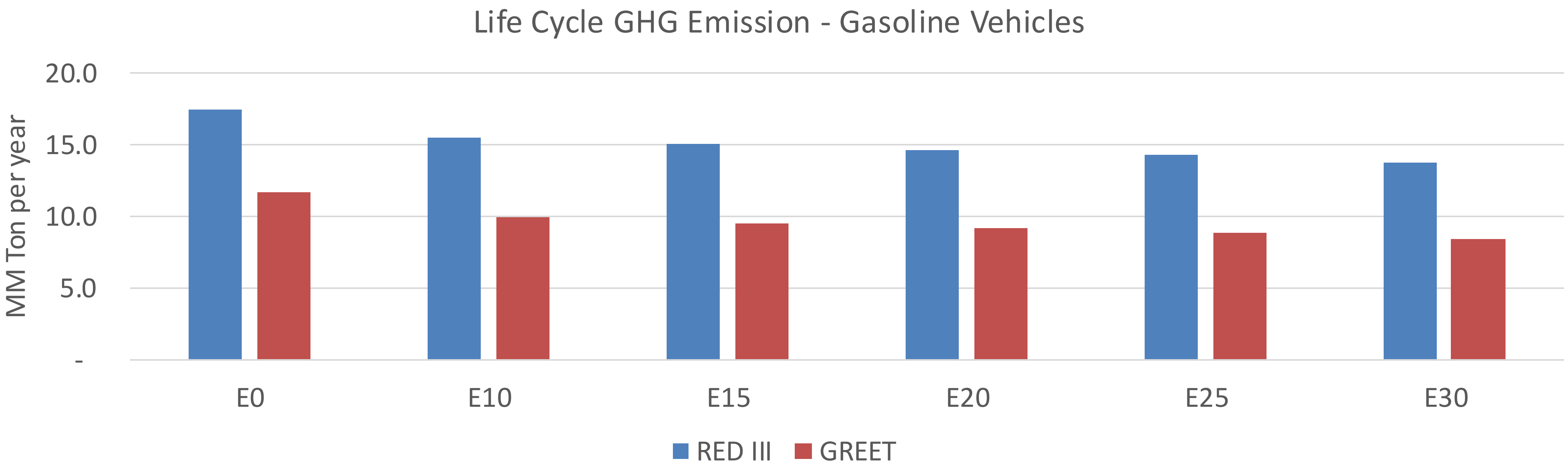
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Czechia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	108.0
Target @2035 (MMT/year)	60.8
Delta	47.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	15.1%	18.6%	21.5%	24.3%	27.9%
RED II/III	0.0%	11.1%	13.8%	16.2%	18.4%	21.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.8%	4.6%	5.3%	6.0%	6.9%
RED II/III	0.0%	4.1%	5.1%	6.0%	6.8%	7.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90